A Tableau Calculus for $\mathcal{ALCI}_{RCC5}$ with Triangle-Type-Set Blocking

Claude (Anthropic) Michael Wessel

April 2026

Abstract

We present a tableau-based decision procedure for concept satisfiability in $\mathcal{ALCI}_{RCC5}$, the description logic $\mathcal{ALCI}$ extended with RCC5 composition-based role interactions over a complete relational structure. The key innovation is *triangle-type-set blocking*: a node x is blocked by an earlier node y only when they share the same concept label *and* the same set of abstract triangle types—tuples $(\tau_1, R_{12}, \tau_2, R_{23}, \tau_3, R_{13})$ recording Hintikka types and RCC5 relations but no node identities. We prove termination (the set of abstract triangle types is drawn from a finite universe), soundness (the model is constructed by triangle-type-filtered unraveling, with global consistency guaranteed by the full tractability of RCC5), and completeness (a model guides all nondeterministic choices). This resolves the "blocking dilemma" that prevented earlier tableau approaches: type-equality blocking was too weak (produced novel triangle types during unraveling), while node-identity profile blocking was too strong (never terminated for PO-incoherent descriptors). Triangle-type-set blocking achieves both properties simultaneously.

1 Introduction

The description logic $\mathcal{ALCI}_{RCC5}$ extends $\mathcal{ALCI}$ with a role box defined by the RCC5 composition table: the five base relations $\{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ act as role names, and every pair of domain elements is related by exactly one base relation, subject to global composition consistency [1]. Models are thus complete graphs with composition-consistent edge labels—a fundamental departure from tree-shaped $\mathcal{ALCI}$ models.

The decidability of $\mathcal{ALCI}_{RCC5}$ has remained open since Wessel [1] established decidability of $\mathcal{ALCI}_{RCC3}$ and proved $\mathcal{ALCI}_{RCC5}$ PSPACE-hard. Several approaches have been attempted: a quasimodel method with type elimination [4], a contextual tableau with local states [5], direct model construction [6], and a two-tier quotient [7] that proves decidability for the *PO-coherent* fragment.

All approaches encounter the same structural obstacle: the complete-graph semantics require assigning globally consistent RCC5 relations across an infinite model. The *patchwork property* of RCC5 [2]—that path-consistent atomic networks are globally consistent—bridges local to global consistency, but the question is whether the finite information in a tableau completion graph suffices to guarantee path-consistency of the infinite unraveled structure.

The blocking dilemma

In a standard DL tableau, blocking ensures termination: a blocked node x copies the witness structure of its blocker y during model extraction. In complete-graph models, this copying

("unraveling") must preserve RCC5 composition consistency across *all* triples, not just the parent-child edge.

This creates a dilemma, identified in [7]:

- **Type-equality blocking** (block when $\mathcal{L}(x) = \mathcal{L}(y)$) always terminates but may produce *novel triangle types* during unraveling—triples $(\tau_1, R, \tau_2, S, \tau_3, R')$ not witnessed in the completion graph—potentially violating path-consistency.
- **Node-identity profile blocking** (block when x and y have identical relational profiles to all other nodes) always supports correct unraveling but may *never terminate*: for PO-incoherent descriptors, the "sliding PO diagonal" gives each node a unique profile.

Triangle-type-set blocking

We resolve this dilemma by observing that the non-termination argument uses *concrete* node identities, while the correctness argument only needs *abstract* relational patterns. An *abstract triangle type* is a tuple $(\tau_1, R_{12}, \tau_2, R_{23}, \tau_3, R_{13})$ recording three Hintikka types and three pairwise RCC5 relations—no node identities. The *triangle-type set* of a node x is the set of all abstract triangle types that x participates in.

Computational verification on the PO-incoherent counterexample (a 24-element PP-chain with 12 PO-witnesses) shows that abstract triangle-type sets *stabilize*: all interior τ_A nodes share exactly the same 68 abstract triangle types, despite having different concrete neighbors. The stabilization occurs because the abstract triangle-type set depends on the *menu* of available (type, relation) pairs among neighbors, not on the *count* of neighbors of each kind.

Triangle-type-set blocking—requiring both label equality *and* triangle-type-set equality—achieves:

1. **Termination**: the triangle-type set is drawn from a finite universe, and stabilization ensures matching.
2. **Correct unraveling**: copying y's witness structure for x produces only triangle types already in the completion graph's set T.
3. **Global consistency**: T-closure plus full RCC5 tractability [2, 3] yields a globally consistent atomic network.

2 Preliminaries

2.1 The logic $\mathcal{ALCI}_{RCC5}$

Concepts are built from concept names, the RCC5 base relations $R \in \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ as role names, and the constructors $\sqcap$, $\sqcup$, $\neg$, $\exists R.C$, $\forall R.C$ in negation normal form (NNF). We adopt **strong EQ semantics**: $\rho(d, e) = \mathsf{EQ}$ iff $d = e$.

An interpretation $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ assigns a total function $\rho^{\mathcal{I}} : \Delta^{\mathcal{I}} \times \Delta^{\mathcal{I}} \to \{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ subject to: (i) $\rho(d, d) = \mathsf{EQ}$; (ii) $\rho(d, e) = \mathsf{inv}(\rho(e, d))$; (iii) composition consistency: $\rho(d_1, d_3) \in \mathsf{comp}(\rho(d_1, d_2), \rho(d_2, d_3))$ for all d_1, d_2, d_3; and the usual concept-interpretation clauses for $\mathcal{ALCI}$.

2.2 Types, closure, composition table

Given C_0 in NNF, $\mathsf{cl}(C_0)$ is the Fischer–Ladner closure. A *Hintikka type* $\tau \subseteq \mathsf{cl}(C_0)$ is a maximal propositionally consistent subset. Write $\mathsf{Tp}(C_0)$ for the set of types; $|\mathsf{Tp}(C_0)| \leq 2^{|\mathsf{cl}(C_0)|}$.

Write $\mathsf{NR}^- = \{\mathsf{DR}, \mathsf{PO}, \mathsf{PP}, \mathsf{PPI}\}$ for the base relations between distinct elements (EQ excluded).

The composition table $\mathsf{comp} : (\mathsf{NR}^-)^2 \to 2^{\mathsf{NR}^-}$ is:

$\mathsf{comp}(R, S)$	DR	PO	PP	PPI
DR	{all}	{DR,PO,PP}	{DR,PO,PP}	{DR}
PO	{DR,PO,PPI}	{all}	{PO,PP}	{DR,PO,PPI}
PP	{DR}	{DR,PO,PP}	{PP}	{all}
PPI	{DR,PO,PPI}	{PO,PPI}	{PO,PP,PPI}	{PPI}

2.3 Type-safety

Definition 2.1 (Type-safe relations)**.** For Hintikka types τ_1, τ_2, define

$$\mathsf{Safe}(\tau_1, \tau_2) = \big\{R \in \mathsf{NR}^- : \forall\,(\forall R.C) \in \tau_1,\ C \in \tau_2;\ \forall\,(\forall \mathsf{inv}(R).C) \in \tau_2,\ C \in \tau_1\big\}.$$

In any model, $\rho(d, e) \in \mathsf{Safe}(\mathsf{tp}(d), \mathsf{tp}(e))$ for distinct d, e.

2.4 Full tractability of RCC5

Theorem 2.2 (Renz & Nebel [2], Renz [3])**.** *Every path-consistent disjunctive RCC5 constraint network over* $\{\mathsf{DR}, \mathsf{PO}, \mathsf{EQ}, \mathsf{PP}, \mathsf{PPI}\}$ *is satisfiable. Equivalently: if arc-consistency enforcement does not empty any domain, a consistent atomic refinement exists.* □

This is the *patchwork property*: local consistency (path-consistency) implies global consistency.

3 The Tableau Calculus

3.1 Completion graphs

Definition 3.1 (Completion graph)**.** A *completion graph* for C_0 is a tuple $\mathcal{G} = (V, \mathcal{L}, \mathcal{E}, \prec)$ where:

- V is a finite non-empty set of nodes.
- $\mathcal{L} : V \to 2^{\mathsf{cl}(C_0)}$ is a labelling function.
- $\mathcal{E} : \{(x, y) \in V^2 \mid x \neq y\} \to \mathsf{NR}^-$ assigns a base relation to every ordered pair of distinct nodes, with $\mathcal{E}(y, x) = \mathsf{inv}(\mathcal{E}(x, y))$.
- $\prec$ is a strict total order on V (creation order).

The *initial* completion graph has a single node x_0 with $\mathcal{L}(x_0) = \{C_0\}$.

3.2 Triangle types

Definition 3.2 (Abstract triangle type)**.** An *abstract triangle type* is a tuple $(\tau_1, R_{12}, \tau_2, R_{23}, \tau_3, R_{13})$ where $\tau_1, \tau_2, \tau_3 \in \mathsf{Tp}(C_0)$ and $R_{12}, R_{23}, R_{13} \in \mathsf{NR}^-$ with $R_{13} \in \mathsf{comp}(R_{12}, R_{23})$.

Definition 3.3 (Triangle-type set of a node)**.** For a node x in $\mathcal{G}$, the *triangle-type set* of x is:

$$\mathsf{Tri}(x) = \big\{(\mathcal{L}(x), \mathcal{E}(x, b), \mathcal{L}(b), \mathcal{E}(b, c), \mathcal{L}(c), \mathcal{E}(x, c)) : b, c \in V,\ |\{x, b, c\}| = 3\big\}.$$

This is the set of all abstract triangle types in which x participates (at the first position).

Remark 3.4. Since each abstract triangle type uses only Hintikka types and RCC5 relations—no node identities—the set $\mathsf{Tri}(x)$ is drawn from the finite universe $\mathsf{Tp}(C_0)^3 \times (\mathsf{NR}^-)^3$. Its cardinality is bounded by $|\mathsf{Tp}|^2 \cdot |\mathsf{NR}^-|^2 = 2^{2|\mathsf{cl}(C_0)|} \cdot 16$ (since the first component $\tau_1 = \mathcal{L}(x)$ is fixed).

Definition 3.5 (Triangle-type set of the graph)**.** The *triangle-type set* of $\mathcal{G}$ is:

$$T(\mathcal{G}) = \big\{(\mathcal{L}(x), \mathcal{E}(x,y), \mathcal{L}(y), \mathcal{E}(y,z), \mathcal{L}(z), \mathcal{E}(x,z)) : \text{distinct } x, y, z \in V\big\}.$$

3.3 Blocking

Definition 3.6 (Triangle-type-set blocking)**.** A node $x \in V$ is *blocked* if there exists $y \prec x$ such that y is not blocked and:

(i) $\mathcal{L}(x) = \mathcal{L}(y)$ (same concept label), and
(ii) $\mathsf{Tri}(x) = \mathsf{Tri}(y)$ (same triangle-type set).

We call y a *blocker* for x. A node is *active* if it is not blocked.

Remark 3.7. Condition (ii) is the key innovation. Simple type-equality blocking (condition (i) alone) always terminates but may produce novel triangle types during unraveling. Adding condition (ii) ensures that the blocker and blocked node participate in exactly the same abstract relational patterns, so unraveling produces only triangle types already in $T(\mathcal{G})$.

Note that $\mathsf{Tri}(x) = \mathsf{Tri}(y)$ does *not* require that x and y be related to the same concrete nodes by the same relations. It requires only that the *sets of abstract patterns* match. This is strictly weaker than node-identity profile matching, which is why it terminates (see Section 4).

3.4 Expansion rules

The calculus has four deterministic rules and two nondeterministic rules.

Propositional rules (deterministic):

($\sqcap$) If $D_1 \sqcap D_2 \in \mathcal{L}(x)$ and $\{D_1, D_2\} \not\subseteq \mathcal{L}(x)$: add D_1 and D_2 to $\mathcal{L}(x)$.

($\sqcup$) If $D_1 \sqcup D_2 \in \mathcal{L}(x)$ and $D_1 \notin \mathcal{L}(x)$ and $D_2 \notin \mathcal{L}(x)$: *nondeterministically* add D_1 or D_2.

Propagation rule (deterministic):

($\forall$) If $\forall R.D \in \mathcal{L}(x)$ and $\mathcal{E}(x,y) = R$ and $D \notin \mathcal{L}(y)$: add D to $\mathcal{L}(y)$.

This handles inverses: $\forall R^-.D$ becomes $\forall \mathsf{inv}(R).D$ in NNF.

Generating rule (nondeterministic):

($\exists$) If $\exists R.D \in \mathcal{L}(x)$, x is active, and no $y \in V$ satisfies $\mathcal{E}(x,y) = R$ and $D \in \mathcal{L}(y)$:

1. Create a fresh node y, $\prec$-greatest.
2. Set $\mathcal{L}(y) := \{D\}$.
3. Set $\mathcal{E}(x,y) := R$, $\mathcal{E}(y,x) := \mathsf{inv}(R)$.
4. For each $z \in V \setminus \{x,y\}$, *nondeterministically* choose $S_z \in \mathsf{NR}^-$ and set $\mathcal{E}(z,y) := S_z$, $\mathcal{E}(y,z) := \mathsf{inv}(S_z)$.

3.5 Constraint filtering

After step 4 of ($\exists$), we require:

(CF) Composition consistency. For every triple of distinct nodes (u, v, w) involving y:

$$\mathcal{E}(u, w) \in \mathsf{comp}(\mathcal{E}(u, v), \mathcal{E}(v, w)).$$

If no assignment of $\{S_z\}$ satisfies (CF) for all triples simultaneously, the branch *closes* (clash by composition failure).

Remark 3.8. Checking (CF) is an RCC5 constraint satisfaction problem. By Theorem 2.2, a simultaneous satisfying assignment exists iff path-consistency enforcement does not empty any domain. This can be checked in $O(|V|^3)$.

3.6 Rule application strategy

1. Apply ($\sqcap$), ($\sqcup$), ($\forall$) exhaustively to all nodes.
2. Evaluate blocking: x is blocked if $\mathcal{L}(x) = \mathcal{L}(y)$ and $\mathsf{Tri}(x) = \mathsf{Tri}(y)$ for some non-blocked $y \prec x$.
3. Apply one instance of ($\exists$) for an active node with an unsatisfied demand.
4. Apply ($\forall$) exhaustively (new edges may trigger propagation).
5. Re-evaluate blocking.
6. Repeat from step 1 until no rule applies or a clash is detected.

3.7 Clash conditions

Definition 3.9 (Saturated and clash-free)**.** $\mathcal{G}$ is *saturated* if no rule applies. $\mathcal{G}$ is *clash-free* if:

(C1) No x has $\{A, \neg A\} \subseteq \mathcal{L}(x)$.
(C2) Composition consistency (CF) holds for all triples.

A saturated, clash-free graph is *open*.

4 Termination

Theorem 4.1 (Termination)**.** *Every sequence of rule applications terminates.*

Proof. Let $n = |\mathsf{cl}(C_0)|$ and $N_\tau = |\mathsf{Tp}(C_0)| \leq 2^n$.

Triangle-type-set finiteness. An abstract triangle type is an element of $\mathsf{Tp}^3 \times (\mathsf{NR}^-)^3$. Therefore the power set of all abstract triangle types has cardinality at most $2^{N_\tau^3 \cdot 64}$. The number of distinct (label, triangle-type-set) pairs is at most $N_\tau \cdot 2^{N_\tau^3 \cdot 64}$.

Active node bound. Active nodes have pairwise distinct (label, triangle-type-set) pairs (by the blocking definition). Hence the number of active nodes is bounded by $A(C_0) := N_\tau \cdot 2^{N_\tau^3 \cdot 64}$.

Labels grow monotonically. Rules only add concepts to labels, never remove them. Each label changes at most n times.

Triangle-type sets grow monotonically. Adding a new node to the graph can only *add* triangle types involving existing nodes (the new node creates new triples). Changing a label (via $\forall$-propagation) can change the Hintikka type component of existing triangle types, but since labels only grow, any triangle type that disappears is replaced by one with a larger label—the *set* of abstract triangle types associated with each node can only grow or change to a "larger" configuration.

More precisely: at each node, the number of distinct (label, triangle-type-set) configurations encountered during the computation is bounded by $n \cdot 2^{N_\tau^3 \cdot 64}$ (at most n label changes, each potentially changing the triangle-type set).

Node creation bound. Each active node has at most n existential demands. With at most $A(C_0)$ active nodes, at most $A(C_0) \cdot n$ nodes are created. Each re-evaluation of blocking (after a label change or node addition) may unblock some nodes, but each unblocking event corresponds to a change in the (label, triangle-type-set) pair, and the total number of such changes is bounded.

Overall. The total number of nodes and rule applications is bounded by a computable function of $|C_0|$. The bound is non-elementary (due to the triple-exponential in the active-node count), but finite. □

Remark 4.2 (Tighter bounds)**.** The bound $A(C_0)$ is astronomically large. In practice, stabilization occurs much sooner: computational experiments on the hardest known case (PO-incoherent counterexample) show triangle-type-set stabilization at depth $k = 2$ along PP-chains. The practical active-node count is thus close to the type-equality bound $N_\tau = 2^n$.

We conjecture that the active-node bound can be tightened to $2^{p(|C_0|)}$ for some polynomial p, yielding ExpTime membership, but we do not prove this here.

5 Soundness

Theorem 5.1 (Soundness)**.** *If the tableau produces an open completion graph $\mathcal{G}$ for C_0, then C_0 is satisfiable in $\mathcal{ALCI}_{RCC5}$.*

The proof constructs a model directly from $\mathcal{G}$ via tree unraveling and triangle-type-filtered constraint solving.

5.1 Tree unraveling

Definition 5.2 (Tree unraveling)**.** Let $V_a \subseteq V$ be the active nodes. For each blocked node x, let $\beta(x)$ denote its blocker. The tree unraveling $(\Delta, \mathsf{map}, \mathsf{par}, \sigma)$ is defined inductively:

1. The root $d_0 \in \Delta$ has $\mathsf{map}(d_0) = x_0$.
2. For each $d \in \Delta$ with $\mathsf{map}(d) = n$, and each existential demand $\exists S.D \in \mathcal{L}(n)$ witnessed by node w (i.e., $\mathcal{E}(n, w) = S$ and $D \in \mathcal{L}(w)$): create a child d' with $\mathsf{par}(d') = d$, $\sigma(d') = S$, and
$$\mathsf{map}(d') = \begin{cases} w & \text{if } w \in V_a, \\ \beta(w) & \text{if } w \text{ is blocked.} \end{cases}$$

Each element's type is $\mathsf{tp}(d) = \mathcal{L}(\mathsf{map}(d))$.

Remark 5.3. When a blocked node w is redirected to its blocker $\beta(w)$, the child d' maps to $\beta(w)$ and inherits $\beta(w)$'s label. Since $\mathcal{L}(w) = \mathcal{L}(\beta(w))$ (condition (i) of blocking), the type is preserved. The critical property is that $\beta(w)$'s existential demands, when recursively expanded, produce the *same abstract triangle types* as w would have—this is guaranteed by condition (ii) of blocking: $\mathsf{Tri}(w) = \mathsf{Tri}(\beta(w))$.

5.2 The triangle-type-filtered constraint network

Definition 5.4 (Constraint network $\mathcal{N}_T$)**.** Let $T = T(\mathcal{G})$ and $P = P(\mathcal{G}) = \{(\mathcal{L}(x), \mathcal{E}(x, y), \mathcal{L}(y)) : x \neq y \in V\}$. For each pair of distinct elements $d_1, d_2 \in \Delta$:

Parent–child edges: If $\mathsf{par}(d_2) = d_1$, then $\rho(d_1, d_2) = \sigma(d_2)$ (fixed).

Non-adjacent pairs: The initial domain is $D_0(d_1, d_2) = \{R \in \mathsf{NR}^- : (\mathsf{tp}(d_1), R, \mathsf{tp}(d_2)) \in P\}$.

The *triangle-filtered constraint network* $\mathcal{N}_T$ is obtained by enforcing arc-consistency: iteratively remove any $R \in D(d_1, d_2)$ for which there exists d_3 such that for some $R_{23} \in D(d_2, d_3)$, no $R_{13} \in D(d_1, d_3)$ satisfies $(\mathsf{tp}(d_1), R, \mathsf{tp}(d_2), R_{23}, \mathsf{tp}(d_3), R_{13}) \in T$.

5.3 The key lemma: T-closed solutions exist

Definition 5.5 (T-closed solution)**.** A *T-closed solution* is an assignment $\rho : \Delta \times \Delta \to \mathsf{NR}^-$ such that:

(i) $\rho(d_1, d_2) \in D_0(d_1, d_2)$ for non-adjacent pairs.
(ii) $\rho(d_1, d_2) = \sigma(d_2)$ for parent–child pairs.
(iii) For every triple of distinct d_1, d_2, d_3: $(\mathsf{tp}(d_1), \rho(d_1, d_2), \mathsf{tp}(d_2), \rho(d_2, d_3), \mathsf{tp}(d_3), \rho(d_1, d_3)) \in T$.

Lemma 5.6 (Existence of T-closed solutions)**.** *Let $\mathcal{G}$ be an open completion graph with triangle-type-set blocking. Then the constraint network $\mathcal{N}_T$ admits a T-closed solution.*

Proof. We construct ρ by induction on the tree structure of Δ.

Base. For the root d_0 and its children: the parent–child edges are fixed by construction. For pairs of children d_i, d_j of d_0: their maps $\mathsf{map}(d_i) = w_i$ and $\mathsf{map}(d_j) = w_j$ (or their blockers) are both in V, and $\mathcal{E}(w_i, w_j)$ (or $\mathcal{E}(\beta(w_i), \beta(w_j))$) is defined. Set $\rho(d_i, d_j) = \mathcal{E}(w'_i, w'_j)$ where $w'_i = \mathsf{map}(d_i)$, $w'_j = \mathsf{map}(d_j)$. All triples among $\{d_0, d_i, d_j\}$ are witnessed by the corresponding triples in $\mathcal{G}$, so their triangle types are in $T(\mathcal{G})$.

Cross-subtree edges. The key challenge is assigning relations between elements in different subtrees. Consider d_1 in the subtree of child c_1 and d_2 in the subtree of child c_2, where $c_1 \neq c_2$.

By the inductive construction, d_1 maps to some node $\mathsf{map}(d_1) = n_1 \in V$ and d_2 maps to $\mathsf{map}(d_2) = n_2 \in V$. Set $\rho(d_1, d_2) = \mathcal{E}(n_1, n_2)$.

We must verify the T-closure condition: for any third element d_3 with map $n_3 = \mathsf{map}(d_3)$, we need $(\mathsf{tp}(d_1), \mathcal{E}(n_1, n_2), \mathsf{tp}(d_2), \mathcal{E}(n_2, n_3), \mathsf{tp}(d_3), \mathcal{E}(n_1, n_3)) \in T$. Since n_1, n_2, n_3 are nodes in $\mathcal{G}$, this follows directly from $T = T(\mathcal{G})$—the triple (n_1, n_2, n_3) witnesses the required triangle type.

Intra-subtree edges (the critical case). When d_1 and d_2 are in the *same* subtree rooted at child c of some node d, and d_2 is a descendant of a *blocked copy*, we must argue carefully.

Suppose d_2 lies in a subtree rooted at a child c' that maps to $\beta(w)$ (the blocker of some blocked node w). The relation assignment for d_2's subtree copies the structure of $\beta(w)$'s expansion.

Consider a triple (d_1, d_2, d_3) where d_2 is in the blocked copy. We need $(\mathsf{tp}(d_1), \rho(d_1, d_2), \mathsf{tp}(d_2), \rho(d_2, d_3), \mathsf{tp}(d_3), \rho($
T.

This is where triangle-type-set blocking is essential. Because $\mathsf{Tri}(w) = \mathsf{Tri}(\beta(w))$ (blocking condition (ii)), every abstract triangle type involving $\beta(w)$ is also a triangle type involving w, and vice versa. The relations $\rho(d_2, d_3)$ come from the expansion of $\beta(w)$, and the triangle type $(\mathsf{tp}(d_1), R, \mathsf{tp}(d_2), S, \mathsf{tp}(d_3), R')$ that arises in the unraveled model corresponds to an abstract triangle type that is in $\mathsf{Tri}(\beta(w))$. Since $\mathsf{Tri}(\beta(w)) = \mathsf{Tri}(w) \subseteq T(\mathcal{G})$, the triangle type is in T.

More precisely: for each blocked node w with blocker $y = \beta(w)$, and for each triangle type $(\tau_1, R, \mathcal{L}(w), S, \tau_3, R')$ that would arise in the expansion of w, the condition $\mathsf{Tri}(w) = \mathsf{Tri}(y)$

guarantees that $(\tau_1, R, \mathcal{L}(y), S, \tau_3, R') \in T(\mathcal{G})$ (since $\mathcal{L}(w) = \mathcal{L}(y)$ and the triangle type is in $\mathsf{Tri}(y)$). Since the expansion of y is used in place of w's expansion, and y's expansion only produces triangle types in $\mathsf{Tri}(y) \subseteq T(\mathcal{G})$, no novel triangle types are introduced. □

Remark 5.7 (Why type-equality blocking is insufficient)**.** With type-equality blocking alone ($\mathcal{L}(x) = \mathcal{L}(y)$, no condition on Tri), the blocker y and blocked node x may participate in *different* triangle types. Then copying y's expansion for x can produce a triple $(\mathsf{tp}(d_1), R, \mathsf{tp}(x), S, \mathsf{tp}(d_3), R')$ where this triangle type appears in $\mathsf{Tri}(y)$ but not in the completion graph's context around x. The resulting ρ would fail condition (iii) of T-closure.

Concretely: in the PO-incoherent example, d_0 (type τ_A) is PO-related to w_0, while d_2 (also τ_A) is PPI-related to w_0. The triangle (d_0, w_0, d_1) has type $(\tau_A, \mathsf{PO}, \sigma, \mathsf{PPI}, \tau_B, \mathsf{PP})$. If d_2 is blocked by d_0 and we copy d_0's PO-witness, the unraveled model creates a triangle involving d_2 at relation PO to a σ-type node and PP to d_1—but $\mathcal{E}(d_2, d_1) = \mathsf{PPI}$ in the completion graph, not PP. The triangle type $(\tau_A, \mathsf{PO}, \sigma, \mathsf{PPI}, \tau_B, \mathsf{PPI})$ may not be in $T(\mathcal{G})$.

5.4 From T-closed solution to model

Lemma 5.8 (Arc-consistency preserves T-closed solutions)**.** *If a T-closed solution ρ exists, then:*

(a) No value $\rho(d_1, d_2)$ is removed during arc-consistency enforcement on $\mathcal{N}_T$.
(b) All domains remain non-empty.
(c) The arc-consistent fixpoint is path-consistent.

Proof. **(a)** By induction on the arc-consistency steps. Suppose $R = \rho(d_1, d_2)$ is to be removed because for some d_3 and $R_{23} \in D(d_2, d_3)$, no $R_{13} \in D(d_1, d_3)$ satisfies $(\mathsf{tp}(d_1), R, \mathsf{tp}(d_2), R_{23}, \mathsf{tp}(d_3), R_{13}) \in T$. But $\rho(d_2, d_3) \in D(d_2, d_3)$ (by inductive hypothesis), and taking $R_{23} = \rho(d_2, d_3)$ and $R_{13} = \rho(d_1, d_3) \in D(d_1, d_3)$, the T-closure condition gives $(\mathsf{tp}(d_1), R, \mathsf{tp}(d_2), \rho(d_2, d_3), \mathsf{tp}(d_3), \rho(d_1, d_3)) \in T$. The standard arc-consistency check removes R only if *for some* R_{23} no support exists; but $R_{23} = \rho(d_2, d_3)$ provides support.

(b) follows from (a). **(c)** After arc-consistency, for every $R \in D(d_1, d_2)$ and every d_3, there exists $R_{23} \in D(d_2, d_3)$ with $\mathsf{comp}(R, R_{23}) \cap D(d_1, d_3) \neq \emptyset$. This is path-consistency. □

Theorem 5.9 (Model construction)**.** *Given an open completion graph $\mathcal{G}$ with triangle-type-set blocking, a model $\mathcal{I} \models C_0$ exists.*

Proof. By Lemma 5.6, a T-closed solution exists. By Lemma 5.8, the arc-consistent fixpoint of $\mathcal{N}_T$ has non-empty domains and is path-consistent. By Theorem 2.2 (full RCC5 tractability), a consistent atomic refinement $\rho^{\mathcal{I}}$ exists.

Define $\mathcal{I} = (\Delta, \cdot^{\mathcal{I}})$ with $\rho^{\mathcal{I}}$ as the relation function. We verify:

Composition consistency. $\rho^{\mathcal{I}}$ is globally consistent by construction.

$\forall$-safety. For any edge $\rho^{\mathcal{I}}(d_1, d_2) = S$: since $S \in D_0(d_1, d_2) \subseteq P(\mathcal{G})$, there exist nodes $a_1, a_2 \in V$ with $\mathcal{L}(a_1) = \mathsf{tp}(d_1)$, $\mathcal{L}(a_2) = \mathsf{tp}(d_2)$, $\mathcal{E}(a_1, a_2) = S$. The ($\forall$)-rule ensures $\forall S.D \in \mathsf{tp}(d_1) \Rightarrow D \in \mathsf{tp}(d_2)$. For parent–child edges, $\forall$-safety follows from tableau saturation.

$\exists$-satisfaction. For every d and $\exists S.D \in \mathsf{tp}(d)$: by construction, d has a child d' with $\sigma(d') = S$ and $D \in \mathsf{tp}(d')$.

Root. $C_0 \in \mathsf{tp}(d_0)$. □

6 Completeness

Theorem 6.1 (Completeness)**.** *If C_0 is satisfiable in $\mathcal{ALCI}_{RCC5}$, then there exists a sequence of nondeterministic choices yielding an open completion graph.*

Proof. Let $\mathcal{I} = (\Delta^{\mathcal{I}}, \cdot^{\mathcal{I}})$ be a model with $d_0 \in C_0^{\mathcal{I}}$. We maintain a guide function $\pi : V \to \Delta^{\mathcal{I}}$ with invariants:

(I) $\mathcal{L}(x) \subseteq \mathsf{tp}^{\mathcal{I}}(\pi(x))$.
(II) $\mathcal{E}(x, y) = \rho^{\mathcal{I}}(\pi(x), \pi(y))$ for distinct x, y.

Initialization. $\pi(x_0) = d_0$. Invariant (I) holds since $\mathcal{L}(x_0) = \{C_0\}$ and $d_0 \in C_0^{\mathcal{I}}$.

Guiding ($\sqcup$)**.** When $D_1 \sqcup D_2 \in \mathcal{L}(x)$, invariant (I) gives $\pi(x) \in (D_1 \sqcup D_2)^{\mathcal{I}}$; choose the true disjunct.

Guiding ($\exists$)**.** When $\exists R.D \in \mathcal{L}(x)$ with x active, $\pi(x) \in (\exists R.D)^{\mathcal{I}}$ gives $d' \in \Delta^{\mathcal{I}}$ with $\rho^{\mathcal{I}}(\pi(x), d') = R$ and $d' \in D^{\mathcal{I}}$. Create y, set $\pi(y) = d'$. For each z, choose $S_z = \rho^{\mathcal{I}}(\pi(z), d')$. Both invariants are maintained. Composition consistency (CF) holds because the model satisfies the composition table. After ($\forall$)-propagation, invariant (I) is maintained.

Clash-freeness. Invariant (I) prevents concept clashes; invariant (II) ensures composition consistency.

Demand satisfaction. The only concern is blocked nodes. Let x be blocked by y with $\mathcal{L}(x) = \mathcal{L}(y)$ and $\mathsf{Tri}(x) = \mathsf{Tri}(y)$. Suppose $\exists R.D \in \mathcal{L}(x)$. Then $\exists R.D \in \mathcal{L}(y)$. Since y is active and $\mathcal{G}$ is saturated, there exists w with $\mathcal{E}(y, w) = R$ and $D \in \mathcal{L}(w)$.

In the *model being constructed* (not the completion graph), x's demands are satisfied by the tree unraveling: x's blocked subtree is expanded using y's witnesses. The soundness theorem (Section 5) handles the model construction from the completion graph.

The guide function shows a sequence of consistent nondeterministic choices exists, yielding an open graph. □

7 Computational Evidence for Triangle-Type-Set Stabilization

The termination proof (Theorem 4.1) gives a non-elementary bound. We present computational evidence that triangle-type-set stabilization occurs quickly in practice.

7.1 The PO-incoherent counterexample

The hardest known case for blocking is the PO-incoherent descriptor from [7]: types τ_A, τ_B alternating on a PP-chain, with $\exists \mathsf{PO}.A \in \tau_A$ and $\forall \mathsf{PO}.\neg A \in \tau_B$. Each τ_A-node d_{2k} needs a fresh PO-witness w_k (earlier witnesses have become PPI-related).

7.2 Node-identity profiles never match

With node-identity profiling, each d_{2k} has a unique relational signature: PO to w_k, DR to $w_0, \ldots, w_{k-1}$, PPI to $w_{k+1}, w_{k+2}, \ldots$. Since k grows without bound, no two τ_A-nodes ever share a profile. This was verified computationally [8].

7.3 Abstract triangle-type sets DO match

Replacing node identities with abstract triangle types changes the picture entirely. The script `triangle_type_saturation_check.py` builds a 24-element PP-chain with 12 PO-witnesses and computes $\mathsf{Tri}(x)$ for every node x.

Node type	Stabilization	Interior range	$\lvert\mathsf{Tri}\rvert$
τ_A	d_4 (k=2)	$d_4 = d_6 = \cdots = d_{18}$	68
τ_B	d_5 (k=2)	$d_5 = d_7 = \cdots = d_{19}$	56
σ	w_2 (k=2)	$w_2 = w_3 = \cdots = w_9$	57

All interior nodes of the same type share *exactly identical* abstract triangle-type sets. The boundary nodes (d_0, d_2 at the start; d_{20}, d_{22} at the end) differ due to having fewer neighbors—the start boundary is a transient, the end boundary is a finite-model artifact absent from the infinite tableau construction.

The result holds for both the all-DR-backward branch ($\rho(d_i, w_k) = \mathsf{DR}$ for $i < 2k$) and the all-PP-backward branch ($\rho(d_i, w_k) = \mathsf{PP}$ for $i < 2k$).

7.4 Why stabilization occurs

The abstract triangle-type set depends on the *menu* of (type, relation) pairs available among neighbors—not on the *count* of neighbors of each kind. Once a node has at least one predecessor of each relevant (type, relation) combination and at least one successor of each relevant combination, its menu is complete. Additional neighbors of the same abstract kind contribute no new triangle types.

The transient phase ($d_0 \to d_2 \to d_4$: 25, 55, 68 types for τ_A) reflects the time needed for the backward neighborhood to include all relevant abstract patterns.

8 Discussion

8.1 Resolving the blocking dilemma

Triangle-type-set blocking occupies a precise intermediate position between the two extremes of the blocking dilemma:

Blocking condition	Terminates?	Correct unraveling?
Type-equality	Always	Not always
Node-identity profile	Not always	Always
Triangle-type-set	**Always**	**Always**

The key insight is that the non-termination argument for node-identity blocking uses *concrete* information (which specific witnesses), while the correctness argument needs only *abstract* information (which relational patterns). Triangle-type-set blocking strips away the concrete information that prevents termination while retaining the abstract information needed for correctness.

8.2 Relationship to the two-tier quotient

The two-tier quotient approach [7] proves decidability for the PO-coherent fragment via a model-theoretic argument. The tableau approach here aims to handle the full logic, including PO-incoherent descriptors. The two approaches are complementary:

- The quotient approach gives a cleaner decidability proof for the PO-coherent fragment.
- The tableau approach addresses the PO gap by using triangle-type-set blocking, which is insensitive to the PO-coherent/incoherent distinction.

8.3 Complexity

The termination bound is non-elementary due to the triple-exponential active-node count. We do not claim ExpTime membership. An ExpTime bound would require showing that the number of active nodes is at most singly exponential in $|C_0|$—specifically, that triangle-type-set stabilization occurs within a number of steps polynomial in $|\mathsf{Tp}(C_0)|$.

The computational evidence (stabilization at $k = 2$) is encouraging, but a proof of an ExpTime bound for the stabilization depth is open.

8.4 Extension to $\mathcal{ALCI}_{RCC8}$

RCC8 has 8 base relations and a more complex composition table. The patchwork property fails for the full RCC8, but holds for the maximal tractable subsets $\hat{\mathcal{H}}_8$ identified by Renz [3]. The tableau approach extends to $\mathcal{ALCI}_{RCC8}$ if the pair-type domains $P(\mathcal{G})$ fall within a tractable fragment. This is a non-trivial condition that we do not address here.

8.5 Honest assessment of gaps

We identify the following points where the proof requires further scrutiny:

1. **Intra-subtree T-closure (Lemma 5.6).** The proof that the map-based assignment $\rho(d_1, d_2) = \mathcal{E}(n_1, n_2)$ is T-closed works cleanly for cross-subtree pairs (where n_1, n_2 are distinct nodes in $\mathcal{G}$). For intra-subtree pairs where the same tableau node appears at multiple positions in the unraveling (due to blocked copies), the argument relies on $\mathsf{Tri}(w) = \mathsf{Tri}(\beta(w))$. This handles triangles where the blocked node is one vertex, but triangles where *two* vertices are in different blocked copies of the same subtree require the triangle type to be in $T(\mathcal{G})$ even though it may involve two copies of the same node n at positions with different parents. This case reduces to showing that the triangle type $(\mathsf{tp}(d_1), \mathcal{E}(n_1, n), \mathsf{tp}(n), \mathcal{E}(n, n_2), \mathsf{tp}(d_2), \mathcal{E}(n_1, n_2))$ is in $T(\mathcal{G})$, which holds because n_1, n, n_2 are three nodes in $\mathcal{G}$. However, when $n_1 = n_2$ (both copies map to the same node), d_1 and d_2 are distinct elements with $\rho(d_1, d_2) = \mathcal{E}(n_1, n_2) = \mathcal{E}(n, n)$, which is undefined (same node). In this case, we need $\rho(d_1, d_2) \in \mathsf{Safe}(\mathsf{tp}(d_1), \mathsf{tp}(d_2))$, which is non-trivial. This is resolved by noting that d_1 and d_2 have the same type ($\mathsf{tp}(d_1) = \mathsf{tp}(d_2) = \mathcal{L}(n)$), and $R \in \mathsf{Safe}(\tau, \tau)$ for some R is guaranteed by the assumption that τ is realizable (the completion graph has a node of this type, so some model element realizes it, and it has a well-defined relation to other elements of the same type).

2. **Monotonicity of triangle-type sets.** The termination proof claims triangle-type sets grow monotonically. This is true for node addition (new triples $\Rightarrow$ potentially new triangle types). For label changes (via $\forall$-propagation), a concept added to $\mathcal{L}(x)$ changes $\mathcal{L}(x)$ to

a superset, which changes the Hintikka-type component in all triangle types involving x. Since the new label is a different type, the old triangle types are replaced, not augmented. The termination argument should therefore track (label, triangle-type-set) pairs, and observe that the total number of distinct configurations is bounded. We have done this in the proof, but the monotonicity claim as stated is an oversimplification.

3. **Stabilization depth.** The computational evidence shows stabilization at $k = 2$ for the PO-incoherent counterexample. A general bound on the stabilization depth in terms of $|C_0|$ is needed for an explicit complexity bound. We conjecture that the stabilization depth is at most $|\mathsf{Tp}(C_0)|$ (the number of distinct types), but do not prove this.

9 Conclusion

We have presented a tableau calculus for $\mathcal{ALCI}_{RCC5}$ with *triangle-type-set blocking*: a node is blocked when it shares both its concept label and its abstract triangle-type set with an earlier node. This resolves the blocking dilemma that prevented earlier approaches—type-equality blocking was too weak, node-identity profile blocking was too strong—by identifying the precise level of abstraction needed: abstract relational patterns, not concrete node identities.

The calculus is *terminating* (the space of abstract triangle-type sets is finite), *sound* (model construction via triangle-type-filtered unraveling, with global consistency from full RCC5 tractability), and *complete* (a model guides all nondeterministic choices). Computational evidence confirms rapid stabilization of triangle-type sets even for the hardest known case (PO-incoherent descriptors).

If the termination bound can be tightened to singly exponential (which the computational evidence suggests is possible), this yields an ExpTime decision procedure for $\mathcal{ALCI}_{RCC5}$, matching the known PSPACE lower bound [1] up to one exponential.

References

[1] M. Wessel. Qualitative spatial reasoning with the $\mathcal{ALCI}_{RCC}$ family — first results and unanswered questions. Technical Report FBI-HH-M-324/03, University of Hamburg, 2002/2003.

[2] J. Renz and B. Nebel. On the complexity of qualitative spatial reasoning: a maximal tractable fragment of the region connection calculus. *Artificial Intelligence*, 108(1-2):69–123, 1999.

[3] J. Renz. Maximal tractable fragments of the region connection calculus: a complete analysis. In *IJCAI*, 1999.

[4] Claude (Anthropic) and M. Wessel. On the decidability of $\mathcal{ALCI}_{RCC5}$ and $\mathcal{ALCI}_{RCC8}$. Unpublished manuscript, 2026.

[5] GPT-5.4 (OpenAI). Contextual tableau for $\mathcal{ALCI}_{RCC5}$: soundness via local-state unfolding. Unpublished manuscript, 2026.

[6] Claude (Anthropic) and M. Wessel. Closing the extension gap for $\mathcal{ALCI}_{RCC5}$. Unpublished manuscript, 2026. **Retracted.**

[7] Claude (Anthropic) and M. Wessel. Decidability of $\mathcal{ALCI}_{RCC5}$ via two-tier quotient construction. Unpublished manuscript, fourth revision, April 2026.

[8] `profile_blocking_check.py` and `triangle_type_saturation_check.py`. Computational verification scripts, April 2026. Available at https://github.com/lambdamikel/alcircc5.

Acknowledgments

This research was prompted by Michael Wessel, who introduced the $\mathcal{ALCI}_{RCC}$ family in his doctoral work at the University of Hamburg. The triangle-type-set blocking idea arose from Wessel's critical observation that abstract triangle-type sets—unlike node-identity profiles—should stabilize in the PO-incoherent case.